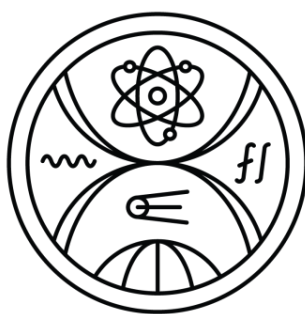


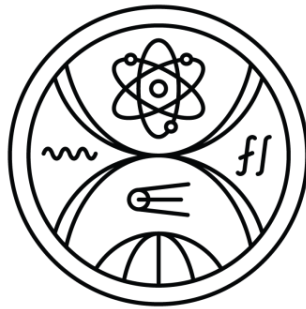
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



IMPLEMENTATION OF AN ABDUCTION
ALGORITHM BASED ON MARCO IN THE CATS
SOLVER

Master thesis

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



IMPLEMENTATION OF AN ABDUCTION ALGORITHM BASED ON MARCO IN THE CATS SOLVER

Master thesis

Study program: Applied computer science
Field of study: Computer science
Department: Department of applied informatics
Supervisor: Mgr. Janka Boborová
Consultant: Mgr. Júlia Pukancová, PhD.



Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

THESIS ASSIGNMENT

Name and Surname: Bc. Anna Viktória Bothová
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Implementation of an abduction algorithm based on MARCO in the CATS solver

Annotation: Abduction is a type of reasoning that explains observed phenomena by identifying their possible causes (explanations) based on available background knowledge. Methods based on HS-trees belong to the most general approaches, enabling a systematic exploration of the explanation space and the step-by-step computation of all possible explanations. These methods require model computation, which is not a service commonly provided by reasoners. Therefore, we aim to explore an alternative approach inspired by the MARCO algorithm, which does not rely on models. Abduction has applications in many domains, including ontology engineering, medical and system diagnosis, multimedia interpretation, and more.

Aim:

1. Analysis of the state of the art, including the existing abduction tool CATS, the principles of HS-tree-based methods, and the incomplete MergeXplain method.
2. Design and modular implementation of an abduction algorithm inspired by the MARCO algorithm and incorporating the MergeXplain method.
3. Optimisation of the implemented abduction algorithm.
4. Empirical evaluation of the implemented algorithm and its comparison with HS-tree-based methods.

Literature:

1. Elsenbroich, C., et al., 2006. A case for abductive reasoning over ontologies. In OWLED. Vol. 216 of CEUR-WS
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4. Shchekotykhin, K. M., et al., 2015. MergeXplain: Fast computation of multiple conflicts for diagnosis. In IJCAI 2015, AAAI Press, pp. 3221–3228.

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ZADANIE ZÁVEREČNEJ PRÁCE

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Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Implementation of an abduction algorithm based on MARCO in the CATS solver
Implementácia nového abdukčného algoritmu založenom na MARCO v nástroji CATS

Anotácia: Abdukcia je typ inferencie, ktorý vysvetľuje pozorované javy určením ich možných príčin (vysvetlení) na základe dostupných znalostí na pozadí. Metódy založené na HS-stromoch patria medzi najvšeobecnejšie prístupy, ktoré umožňujú systematicky prehľadávať priestor vysvetlení a postupne vypočítať všetky možné vysvetlenia. Tieto metódy vyžadujú počítanie modelov, čo nepatrí medzi bežne poskytované služby usudzovačov. Preto sa chceme preskúmať alternatívny prístup inšpirovaný algoritmom MARCO, ktorý modely nevyužíva. Abdukcia nachádza uplatnenie v mnohých oblastiach, vrátane ontologického inžinierstva, medicínskej a systémovej diagnostiky či interpretácie multimédií a ďalších.

Cieľ: 1. Analýza súčasného stavu, vrátane existujúceho abdukčného nástroja CATS, princípu metód založených na HS-stromoch a neúplnej metódy MergeXplain.
2. Návrh a modulárna implementácia abdukčného algoritmu inšpirovaného algoritmom MARCO a využívajúceho metódu MergeXplain.
3. Optimalizácia implementovaného abdukčného algoritmu.
4. Empirická evalvácia implementovaného algoritmu a jeho porovnanie s metódami založenými na HS-stromoch.

Literatúra: 1. Elsenbroich, C., et al., 2006. A case for abductive reasoning over ontologies. In OWLED. Vol. 216 of CEUR-WS
2. Homola, M., et al., 2023. Merge, explain, iterate: A combination of MHS and MXP in an ABox abduction solver. In JELIA. Vol. 14281 of LNCS, Springer Nature
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4. Shchekotykhin, K. M., et al., 2015. MergeXplain: Fast computation of multiple conflicts for diagnosis. In IJCAI 2015, AAAI Press, pp. 3221–3228.

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Dátum schválenia: 13.10.2025

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garant študijného programu

.....
š t u d e n t

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v e d ú c i p r á c e

I hereby declare that I have completed this thesis, including all its attachments and figures, independently, using only the literature listed in the bibliography and artificial intelligence tools. AI tools were used for purposes such as grammatical correction, text refinement, or formatting assistance. I confirm that their use complies with all relevant legal regulations, academic rights and freedoms, ethical and moral standards, and that academic integrity has been fully maintained. The responsibility for the final form and correctness of the text rests solely with the author.

Bratislava, 2026

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Bc. Anna Viktória Bothová

Acknowledgement

Abstract

Keywords:

Abstrakt

Klíčové slová:

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Introduction

Part I

State of the art

Chapter 1

Description logics

Description logics (DLs) are a group of knowledge-representation languages that model the concepts, relationships, and constraints of a domain with precise semantics. They provide a well-organized framework that supports automated reasoning [1]. In this chapter, we will present the syntax, semantics, main reasoning problems in DL, and also describe ontologies.

1.1 Syntax

There is a variety of DLs that differ in their expressive capabilities. The core logic $\mathcal{ALC}$ serves as a baseline, and additional letters are appended to denote the inclusion of further constructors or reasoning features [5]. We illustrate the syntax and semantics of Description Logics using $\mathcal{ALC}$, as it represents one of the smallest yet still expressive and conceptually significant DLs. Description Logics are built upon three mutually disjoint sets of basic elements, which collectively are called *vocabulary* [5]:

Individual names are names assigned to particular, one-of-a-kind entities, such as people, objects, or other items within the domain. Typical instances are peter, emma or rex.

Concept names denote a group of concrete or abstract entities that are defined by shared features. For example, Person, Dog, Warm, or Fragile.

Role names express relations between two entities of a domain, e.g. loves, ownerOf, worksAt.

Concept descriptions, or simply *concepts*, are formed using concept names, role names, and other fundamental components. Two particular concepts are $\top$ (called top concept) and $\perp$ (called bottom concept). Top concept represents the set of all individuals in the domain, it is a superset of all other concepts. Bottom concept is an empty set, it is a subset of all other concepts.

Every concept symbolizes a set of elements in the domain, called its extension. For example, if we know that there exists a statement of the form $\text{Dog}(\text{rex})$, then rex is in the extension of Dog , or shortly, rex is a Dog .

Concepts can be separated into simple and complex (also called compound) forms. Atomic concepts are those denoted by a single lexical unit, in $\mathcal{ALC}$ this includes concept names, $\top$ and $\perp$. A complex concept must contain one or more of the available operators, in $\mathcal{ALC}$ these are the following: conjunction ($\sqcap$), disjunction ($\sqcup$), negation ($\neg$), existential restriction ($\exists$), value restriction ($\forall$).

TODO: write examples

The domain contains statements called axioms. A collection of all axioms in a domain is called *knowledge base*, denoted $\mathcal{K}$. It consists of two main parts. *TBox* is the terminological part, it allows us to define general definitions and constraints about concepts and roles, as well as their relations [2]. The assertional part, expressing the properties and relationships of individuals, is called *ABox*.

Definition 1 (ABox). *Let C be a concept, r a role name, and x, y individual names. An assertial axiom can have the following forms:*

$C(x)$ (concept assertion),

$r(x, y)$ (role assertion),

$\neg r(x, y)$ (negated role assertion).

A collection of concept and role assertions is called *ABox* [1].

TODO: write TBox definition + negated role assertion?? (not in ACL)

1.2 Semantics

After defining the axioms, it is important to understand them. An interpretation, denoted $\mathcal{I}$, is a structure that specifies the meaning of concepts and roles. It has two main parts: an interpretation domain, which is a non-empty set of elements, and a mapping function [5]. The mapping function returns the entity that an individual name refers to. It also establishes the extensions of concept and role names. Thus, it returns a group of entities that are part of a concept name or a pair of elements that have a relationship between them in the form of a specific role.

Definition 2 (Interpretation). *An interpretation is a composition of a set called interpretation domain $\Delta^{\mathcal{I}}$ such that $\Delta^{\mathcal{I}} \neq \emptyset$, and the interpretation function $\cdot^{\mathcal{I}}$, which links the components of the vocabulary to the interpretation domain as follows:*

$a^{\mathcal{I}} \in \Delta^{\mathcal{I}}$,

$A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$,

$r^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$,

$$\begin{aligned}
\top^{\mathcal{I}} &= \Delta^{\mathcal{I}}, \\
\perp^{\mathcal{I}} &= \emptyset, \\
(\neg F)^{\mathcal{I}} &= \Delta^{\mathcal{I}} \setminus F^{\mathcal{I}}, \\
(F \sqcap G)^{\mathcal{I}} &= F^{\mathcal{I}} \cap G^{\mathcal{I}}, \\
(F \sqcup G)^{\mathcal{I}} &= F^{\mathcal{I}} \cup G^{\mathcal{I}}, \\
(\exists r.F)^{\mathcal{I}} &= \{b \in \Delta^{\mathcal{I}} \mid \exists c \in \Delta^{\mathcal{I}} : (b, c) \in r^{\mathcal{I}} \text{ and } c \in F^{\mathcal{I}}\}, \\
(\forall r.F)^{\mathcal{I}} &= \{b \in \Delta^{\mathcal{I}} \mid \forall c \in \Delta^{\mathcal{I}} : \text{ if } (b, c) \in r^{\mathcal{I}} \text{ then } c \in F^{\mathcal{I}}\}
\end{aligned}$$

TODO: axiom satisfaction + model + reasoning, entailment + OWL

1.3 Reasoning problems and models

1.4 Ontologies

Chapter 2

Abduction: theory and algorithms

In this chapter we will introduce abduction and some abductive algorithms.

2.1 Abduction

Classical Description Logic reasoning is almost entirely deductive and monotonic, which makes it poorly suited for tasks that require generating explanations, hypotheses, or diagnoses. Abductive reasoning, which means inferring the best explanation for an observation, fills this gap by enabling ontology-based systems to reason backwards from effects to plausible causes [3].

2.1.1 Abductive reasoning

2.1.2 DL ABox abduction

2.2 Abduction algorithms

There are existing tree-based abduction algorithms. However, they build models, which is computationally expensive. In our new algorithm, we try to omit the need of models and use a new approach. In this chapter, we will introduce the existing algorithms used for abduction and the idea behind our new algorithm and show why it is a new and useful approach.

2.2.1 Conflict based algorithms

2.2.2 HS-tree-based algorithms

2.3 MARCO

The idea behind our new algorithm comes from an article by Liffiton, Previti, Malik, and Marques-Silva, who introduced the idea of computing Minimal Unsatisfiable Subsets (MUSes) of constraint sets. Existing MUS-enumeration techniques often struggle with scalability or rigidity, and they introduce a framework that is both computationally faster and highly adaptable to different constraint domains. Their approach significantly improves the practical feasibility of MUS enumeration, making it a more useful tool for applications that rely on identifying the minimal sources of inconsistency [4].

Chapter 3

CATS solver

In this chapter we will describe the CATS solver, which is used to solve abduction problems with different algorithms. The implementation of our new algorithm takes place in this solver.

3.1 Overview of the CATS solver

what CATS is designed for, what type of problems it solves, why it is relevant to the thesis

3.2 Architecture and core components

classes: TreeBuilder, NodeProcessor

3.3 Algorithms used in CATS

inputs, outputs, algorithms, switches?

Part II

Contribution

Chapter 4

Implementation

In this chapter we will describe the implementation of the new MARCO algorithm.

4.1 Basic MARCO

4.2 Variations of MARCO

Chapter 5

Evaluation

In this chapter, we will analyze the performance of our new MARCO algorithm, compare it to existing abduction algorithms. We will also state the strengths, weaknesses, and possible extensions of this algorithm.

5.1 Dataset and methodology

5.2 Results

Conclusion

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